

Geospatial Data Science Exam Project Instructions, Spring 2026

The submission is a written project report about the application of geospatial data science either to **answer a research question** or to **create a prototype of a digital product**. It may range from a technical workflow proof of concept to research data exploration. The project should explore or solve a problem with a geospatial dimension and may focus on any aspect of spatial data collection, visualization, analysis, or statistical evaluation. The submission has two parts: (a) commented code deposited on Github (or similar code repository) and (b) the associated report that describes the project and links to the code repository. No use of generative artificial intelligence is allowed at the exam under any circumstances.

If anything is left unclear after reading this document, please ask in the forum or in class.

Project approval

Before starting to work on your project, you must form a group of 1, 2, or 3 people, and submit a very short project proposal at the latest **by April 8th** in the Google Sheet linked at LearnIT. All proposals need to be approved by the course manager by April 10th. The reason is to avoid unrealistic plans or substantial overlaps between groups. You will then have 5+ weeks to work on the project until the submission deadline on **May 18th**.

Example data sets and projects

Groups are welcome to collect their own data sets. However, it might be more straightforward to use an existing, publicly available data set. Go here for example data, research, projects, and libraries: <https://learnit.itu.dk/course/section.php?id=168948>

There is no restriction on libraries: Feel free to use whatever works best, as long as you document it.

Project hand-in

You must hand in:

- report.pdf: A project report. See structure, content, and length requirements below.
- gitlog.txt: Your repo's git log, e.g. by running: `git log > gitlog.txt`
- If you hand in data analysis or research: code.zip: One zip file containing Jupyter notebook(s) (.ipynb) of your commented code that runs fully without errors, assuming that all used data files are in place, and reproduces your findings. Do not include the data files here.
- If you hand in a prototype of a digital product, do not submit your code, but provide the link to your code repository in the report. Also provide the URL or a way where your prototype can be tried out.

Report structure, content, and length requirements

Using a layout like in the provided LaTeX/Overleaf template, not including references and appendices, project reports should be:

- 5-9 pages for groups of size 1
- 7-11 pages for groups of size 2
- 9-13 pages for groups of size 3

Your report should contain at least one or more figures that visualize geospatial data. It is up to you whether you want to use the provided LaTeX template or use another one – but using LaTeX or similar is strongly encouraged.

The report should include the motivation and background for your project, a description of

the data and the data processing, your results, a critical discussion of both process and outcome, as well as conclusions and future work (see below for more details).

You are expected to make use of and refer to some of the geospatial concepts and methods we have worked with in the course throughout the report.

You are free to organize your report as you see fit, but the suggested report structure is as follows:

Title or project name, authors

1. Introduction / Goal

Here you provide the context and motivation for the problem. What are your research questions, or what does your prototype intend to do/solve? Explain the particular spatial problem you are tackling and its cultural relevance (please use language accessible to non-specialized but intelligent audience).

2. Problems and Background

- Provide context (historical, cultural, technical) for your project and introduce the background and related work in literature (cite or list relevant literature on the research problem; list other scripts and software in this area etc.)
- Give the formulations of (cultural, historical, social, technical, etc.) problems to be solved by the project and role of the different digital tools in achieving the aims of the project
- Specify your approach

3. Data acquisition and processing

- List and cite all sources of data used in this paper, comment on their fitness for purpose (format, quality, and provenance). Focus on the spatial component of your data, its provenance and precision and reliability.
- Details of data manipulation and transformation. Link to processing scripts where relevant.

4. Results

- Provide and explain the results of your investigation, illustrated with figures where essential and relevant. Relate to lessons learnt, counts, statistics, maps or other outcomes.
- Briefly comment on 1) the main elements of your digital workflow, highlighting challenges and decision-making bottlenecks (e.g. how did you transform point data to make it into a continuous surface?) 2) functions/tricks you found useful and wish to promote or credit. Remember that the technical tasks should not clutter/interfere with your overall narrative and data analysis (unless your project is about developing a technical pipeline)
- For 'technical pipeline' projects: provide a guided tour of your pipeline to facilitate its reproducibility, explaining your choices, clarifying dependencies, and referring to the scripts/tools you compiled in GitHub.

5. Critical evaluation

- Evaluate the results in light of the data sources and research premises/assumptions. How representative, reliable, complete and precise are your results? How transferable or generalizable? Give an account on the major short-coming(s) of your methodology / data / prototype.
- Briefly evaluate the results in light of digital tools, the learning process, time on task, vis-à-vis the final product.

- For 'technical pipeline' projects: Provide a comparison with other state-of-the-art data or software if any exists for the same task (kindly cite relevant work, scripts, etc.)

6. Conclusions and future work

Here you provide a couple of sentences summarizing the results of the project, the achieved (or missed) goals and highlight the most important lessons learnt while working on the project. Indicate how the methods, data, analysis, or prototype functionalities could be improved or extended in future work.

References

At least 5, both domain-based literature and references to digital tutorials or internet resources consulted.

Appendix A – Metadata tables (required)

Table 1 – Software metadata

	Software metadata description	Please fill in this column
S1	Current software version	Example: gds:6.1, Python 3.9
S2	Permanent link to your code in your Github repository	Example: https://github.com/mszell/bikenwgrowth/releases/tag/1.0.0
S3	Legal Software License	Example: Creative Commons 4.0
S4	Computing platform / Operating System	Example: Linux 18.04, OS X, Microsoft Windows 10,...
S5	Installation requirements & dependencies for software not used in class	
S6	If available Link to software documentation for special software	Example: http://mozart.github.io/documentation/
S6	Support email for questions	

Table 2 – Data metadata (use the template below or create your own metadata table)

	Data metadata description	Please fill in this column (you can link to license and metadata descriptions online; where relevant remember to articulate data provenance and quality)
D1	Data License	List the license for your own data, and communicate the license of other used datasets
D2	Dataset name / main properties	Example: Geochem_dk.grid. SpatialRaster layer, resolution 30x30m, attributes: "ID" "Elev" "pH" generated by KU Dept of Geochemistry, 2020 interpolated from 5x5km grid sampling, source: www.xxxx.dk

Appendix B - Contribution statement (mandatory for group sizes above 1)

For individualized grading, here you must clarify who of your group is responsible for the different parts of the project submission. Please state:

- For at least 2 sections of your report, who was primary contributor (and optionally also second and/or third contributor)
- Who was primary (optionally also second and/or third) contributor for any additional major tasks such as data collection, data preparation, analysis, programming, or visualization.